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REPETITION

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Contents

- Repetition construction
- Repetition examples



Repetition

- The computer is able to carry out an instruction repeatedly without getting tired or bored.
- The repetition structure consists of:
  1. Repeat condition; that is, a boolean expression that must be met to carry out the repetition.
  2. Repetition body; i.e. one or more actions/statements that will be repeated.



Repetition structure

- The repetition structure is usually accompanied by these parts:
  1. Initialization; i.e. actions taken before repetition are first performed.
  2. Termination; that is, the action taken so that the repetition is finished.



Repetition structures type

1. FOR
2. WHILE-DO
3. REPEAT-UNTIL (DO-WHILE)



FOR

- The FOR structure is used for the repetition of actions, which are repeated for a certain number.
- The FOR structure can be in the form of:
  1. *Ascending*.
  2. *Descending*.



Ascending FOR

```
for counting_variable ← initial_value to final_value do
      Action/statement1
      Action/statement2
      …
      Action/statementN
endfor
```

- `Counting_variable` must be of the basic type (except real)
- `initial_value` must be smaller or equal to `final_value`
- `Counting_variable` initialized with `initial_value` . `Counting_variable` value increases by one each time the action is repeated until finally, the `Counting_variable` value is the same as the `final_value`
- **Number of repetitions that occur =**

```
final_value - initial_value + 1
```



Example

```
1   PROGRAM PrintingLotsaHello
2   ( printing "HELLO" 10 times )
3
4   DECLARATION:
5       k : integer          { counting_variable }
6
7   ALGORITHM:
8       for k ← 1 to 10 do       { initialization & termination }
9         write('HELLO')
10      endfor
11      { termination condition: k = 10 }
```



Translation

```c
/* PROGRAM PrintingLotsaHello */
/* printing "HELLO" 10 times  */

#include <stdio.h>

main()
{
    /* DECLARATION: */
    int k;      /* counting_variable */

    /* ALGORITHM: */
    for (k = 1; k <= 10; k++)   /* initialization & termination */
    {
        printf("HELLO\n");
    }       /* termination condition: k = 10  */
}
```



Descending FOR

```
for counting_variable ← initial_value downto final_value do
      Action/statement1
      Action/statement2
      …
      Action/statementN
endfor
```

- `Counting_variable` must be of the basic type (except real)
- `initial_value` must be bigger or equal to `final_value`
- `Counting_variable` initialized with `initial_value` . `Counting_variable` value decreases by one each time the action is repeated until finally, the `Counting_variable` value is the same as the `final_value`
- **Number of repetitions that occur =**

  `initial_value - final_value + 1`



Example

```
1   PROGRAM RocketLaunch
2   (Count down the rocket launch by writing to the screen from 100 to
3   0, and then write "Go!")
4
5   DECLARATION:
6       i : integer          { counting_variable }
7
8   ALGORITHM:
9       for i ← 100 downto 0 do      { initialization & termination }
10        write(i)
11      endfor                       { termination condition: i = 0 }
12      write('Go!')
```



Translation

```c
/* PROGRAM RocketLaunch */
/* Count down the rocket launch by writing to the screen from 100 to 0,
and then write "Go!")  */

#include <stdio.h>

main()
{
    /* DECLARATION: */
    int i;      /* counting_variable */

    /* ALGORITHM: */
    for (i = 100; i >= 0; i--)  /* initialization & termination */
    {
        printf("%d\n", i);
    }                           /* termination condition: i = 0  */
    printf("Go!\n");
}
```



WHILE-DO

```
while condition do
        Action/statement1
        Action/statement2
        …
        Action/statementN
endwhile
```



Example

```
1   PROGRAM PrintingLotsaHello
2   ( printing "HELLO" 10 times )
3
4   DECLARATION:
5       k : integer          { counting_variable }
6
7   ALGORITHM:
8       k ← 1                { initialization }
9       while k ≤ 10 do
10        write ('HELLO')
11        k ← k + 1          { termination }
12      endwhile
13      { termination condition: k > 10 }
```



Translation

```c
/* PROGRAM PrintingLotsaHello */
/* printing "HELLO" 10 times  */

#include <stdio.h>

main()
{
    /* DECLARATION: */
    int k;                  /* counting_variable */

    /* ALGORITHM: */
    k = 1;                  /* initialization */
    while (k <= 10)
    {
        printf("HELLO\n");
        k = k + 1;          /* termination */
    }                   /* termination condition: k > 10 */
}
```



REPEAT-UNTIL

```
repeat
        Action/statement1
        Action/statement2
        …
        ActionstatementN
until  condition
```



Example

```
1    PROGRAM PrintingLotsaHello
2    ( printing "HELLO" 10 times )
3
4    DECLARATION:
5        k : integer          { counting_variable }
6
7    ALGORITHM:
8        k ← 1                { initialization }
9        repeat
10         write ('HELLO')
11         k ← k + 1          { termination }
12       until k > 10
13       { termination condition: k > 10 }
```



Translation

```c
/* PROGRAM PrintingLotsaHello */
/* printing "HELLO" 10 times  */

#include <stdio.h>

main()
{
    /* DECLARATION: */
    int k;                  /* counting_variable */

    /* ALGORITHM: */
    k = 1;                  /* initialization */
    do
    {
        printf("HELLO\n");
        k = k + 1;          /* termination */
    } while (k <= 10);  /*termination condition: k > 10 */
}
```



Example

```
1   PROGRAM ProgramMenu
2   ( Program for displaying menus, reading menu options, and displaying
3   user choices )
4
5   DECLARATION:
6       menuNumber : integer          { menu selected }
7
8   ALGORITHM:
9       repeat
10        { printing menus to the screen }
11        write(`       MENU         `)
12        write(` 1. Reading Data    `)
13        write(` 2. Printing Data   `)
14        write(` 3. Updating Data   `)
15        write(` 4. Deleting Data   `)
16        write(` 5. Exit Program    `)          { termination }
17        read(menuNumber)
18
19        case (menuNumber)
20             menuNumber = 1 : write('You picked menu number 1')
21             menuNumber = 2 : write('You picked menu number 2')
22             menuNumber = 3 : write('You picked menu number 3')
23             menuNumber = 4 : write('You picked menu number 4')
24             menuNumber = 5 : write('Exit program')
25        endcase
26
27      until menuNumber = 5
28      { termination condition: menuNumber = 5 }
```



REPEAT-UNTIL example

```c
/* PROGRAM ProgramMenu */
/* Program for displaying menus, reading menu options,
and displaying user choices */

#include <stdio.h>

main()
{
    /* DECLARATION: */
    int menuNumber;     /* menu selected */

    /* ALGORITHM: */
    do
    {
        /* cetak menu ke layar */
        printf("         MENU       \n");
        printf("1. Reading Data     \n");
        printf("2. Printing Data    \n");
        printf("3. Updating Data    \n");
        printf("4. Deleting Data    \n");
        printf("5. Exit Program     \n"); /* termination */
        printf("Please select a menu : "); scanf("%d", &menuNumber);

        switch (menuNumber)
        {
            case 1 : printf("You picked menu number 1\n"); break;
            case 2 : printf("You picked menu number 2\n"); break;
            case 3 : printf("You picked menu number 3\n"); break;
            case 4 : printf("You picked menu number 4\n"); break;
            case 5 : printf("Exit Program\n"); break;
        }
    } while (menuNumber != 5);  /*termination condition: menuNumber = 5 */
}
```



WHILE-DO example

```
1   PROGRAM ProgramMenu
2   ( Program for displaying menus, reading menu options, and displaying
3   user choices )
4
5   DECLARATION:
6       menuNumber : integer          { menu selected }
7
8   ALGORITHM:
9       menuNumber ← 0                { initialization }
10      while (menuNumber ≠ 5)
11        { printing menus to the screen }
12        write('       MENU         ')
13        write(' 1. Reading Data    ')
14        write(' 2. Printing Data   ')
15        write(' 3. Updating Data   ')
16        write(' 4. Deleting Data   ')
17        write(' 5. Exit Program    ')          { termination }
18        read(menuNumber)
19
20        case (menuNumber)
21             menuNumber = 1 : write('You picked menu number 1')
22             menuNumber = 2 : write('You picked menu number 2')
23             menuNumber = 3 : write('You picked menu number 3')
24             menuNumber = 4 : write('You picked menu number 4')
25             menuNumber = 5 : write('Exit program')
26        endcase
27
28      endwhile
29      { termination condition: menuNumber = 5 }
```



REPEAT-UNTIL vs WHILE-DO (?)

```
1   PROGRAM ProgramMenu
2   ( Program for displaying menus, reading menu options, and displaying
3   user choices )
4
5   DECLARATION:
6       menuNumber : integer          { menu selected }
7
8   ALGORITHM:
9       repeat
10        { printing menus to the screen }
11        write(`       MENU        `)
12        write(` 1. Reading Data   `)
13        write(` 2. Printing Data  `)
14        write(` 3. Updating Data  `)
15        write(` 4. Deleting Data  `)
16        write(` 5. Exit Program   `)          { termination }
17        read(menuNumber)
18
19        case (menuNumber)
20              menuNumber = 1 : write('You picked menu number 1')
21              menuNumber = 2 : write('You picked menu number 2')
22              menuNumber = 3 : write('You picked menu number 3')
23              menuNumber = 4 : write('You picked menu number 4')
24              menuNumber = 5 : write('Exit program')
25        endcase
26
27      until menuNumber = 5
28      { termination condition: menuNumber = 5 }
```

```
1   PROGRAM ProgramMenu
2   ( Program for displaying menus, reading menu options, and displaying
3   user choices )
4
5   DECLARATION:
6       menuNumber : integer          { menu selected }
7
8   ALGORITHM:
9       menuNumber ← 0                { initialization }
10      while (menuNumber ≠ 5)
11        { printing menus to the screen }
12        write(`       MENU        `)
13        write(` 1. Reading Data   `)
14        write(` 2. Printing Data  `)
15        write(` 3. Updating Data  `)
16        write(` 4. Deleting Data  `)
17        write(` 5. Exit Program   `)          { termination }
18        read(menuNumber)
19
20        case (menuNumber)
21              menuNumber = 1 : write('You picked menu number 1')
22              menuNumber = 2 : write('You picked menu number 2')
23              menuNumber = 3 : write('You picked menu number 3')
24              menuNumber = 4 : write('You picked menu number 4')
25              menuNumber = 5 : write('Exit program')
26        endcase
27
28      endwhile
29      { termination condition: menuNumber = 5 }
```



Bibliography

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Thank You!



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